

ST 558-651 Course Syllabus

Data Science for Statisticians

Summer 2026

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Course Description

Methods for reading, manipulating, and combining data sources including databases. Custom functions, visualizations, and summaries. Common analyses done by data scientists. Methods for communicating results including dashboards. Regular access to a computer for homework and class exercises is required.

Learning Outcomes

At the end of this course students will be able to

- read, write, and explain the steps and purpose of computer programs
- efficiently read in, combine, and manipulate data
- utilize help and other resources to customize programs
- describe the strengths and weaknesses of the chosen programming language and be able to write efficient programs
- explore data and perform common analyses
- create reports, web pages, and dashboards to display and communicate results

Course Materials

This course engages diverse scholarly perspectives to develop critical thinking, analysis, and debate and inclusion of a reading does not imply endorsement.

Required Textbook

None

Other Required Materials

Software: Students in this course will use the [R statistical software](#) and the [R Studio IDE](#). This software is open source, works on all major platforms, and is free to anyone. It is widely used in statistics and data science (behind python in data science but still quite popular).

Other than off campus proctoring of exams, there are no required costs for students in the course.

Other open source readings are made available on the course website.

Technology Requirements

Access to a computer that you are able to install software on is required for the course.

The prerequisite coursework requires that students have completed a programming course. This implies that students are expected to have a strong understanding of how computers function and the logic required to instruct them.

Software

- [Moodle and Wolfware](#)
 - [Moodle Accessibility Statement](#)
 - [Moodle Privacy Policy](#)
 - [NCSU Privacy Policy](#)
- [Adobe Reader](#) (for reading PDF files)
 - [Accessibility Statement](#)
 - [Adobe Privacy Policy](#)
- [Zoom](#)
 - [Zoom Accessibility Statement](#)
 - [Zoom Privacy Policy](#)
- [G Suite](#)
 - [Accessibility Statement](#)
 - [Privacy Policy](#)
- [Office 365](#)
 - [Accessibility Statement](#)
 - [Privacy Policy](#)
- [R Statistical Software](#) and [R Studio](#) for programming
- [Miktex](#) distribution for creation of PDF files
- [Docker](#) for creating containers
- [Github](#) for version control and collaboration. **We recommend the use of the github rather than NCSU's github.**

NC State University Libraries offers [Technology Lending](#), where many devices are available to borrow for a 7-day period. [Computer labs](#) are available in various locations around campus for student use.



Computer

A computer is required for students taking this course. NC State's Office of Information Technology provides recommendations for [your computer at NC State](#).

Other Devices

A headset with a microphone is recommended for office hours but is not required. Students are recommended to join office hour sessions on their computer with a camera and microphone. If that is not an option, a cell phone may be used.

Other Student Expenses

None

Grading

It is the student's responsibility to be aware of their grades in the course and the appropriate level of work required. The **Moodle gradebook final grades are likely not correct** so be sure to keep track of how you are doing in the course. Your final grade depends on the following (subject to change with notification):

Course Assignment Values

Percentage of grade	Component
15 %	Homework (8)
20 %	Projects (3)
25 %	Exam

Homework: There will be eight homework assignments that give you a low stakes chance to collaboratively practice programming. There will also be some explanation questions included in these to help you practice for the exam. These assignments each have equal weight. **Parts of these**

assignments will not be graded for correctness. Example keys will be posted for each assignment. **It is the student's responsibility to make sure they understand what they did successfully and unsuccessfully on their homework assignments.** Please contact us if you have any questions about your solutions.

Projects: There will be three larger projects with the last project acting as the final for the course. Some of these projects will require collaboration with others. Details will follow as we get closer to the projects being assigned. Grading guides/rubrics will be provided with the assignments.

- **Homework and projects can be turned in by 6:59am the day after they are due with no penalty. (The link stays open until then and you can submit.)**
- Generally, no late work will be accepted - please contact me immediately if you have something unavoidable come up.

Exam: The exam is **closed book and closed notes**. The exam requires an in-person proctor.

Exam window: T-W, 6/23-6/24

Students who are unable to take an exam during the designated exam window for a legitimate unavoidable reason may be given the opportunity for a make-up exam or, in extreme cases, be given a reweighting of their scores. The exam is limited to 75 minutes. Many of the questions will be multiple choice, fill in the blank, or true/false. Other questions will be in essay form. These questions are generally graded on a 0/1 or 0/1/2/3 basis.

Academic Testing Services Information

This course requires proctored exams facilitated through [Academic Testing Services](#). A proctor is an impartial third party who verifies the identity of the student and ensures the academic integrity of an exam.

Local students — Academic Testing Services will offer exams for on-campus students at one of the local test centers. Please visit the Academic Testing Services website for [more information about on-campus testing](#).

- [Step 1: Make an Appointment](#). Exams at the Academic Test Centers are by **appointment only**.
- **Step 2: Come Prepared.**
 - Bring a physical photo ID
 - Know your Unity ID
 - DUO, bring your registered device

Students with Accommodations— If you have approved accommodations with NC State’s Disability Resource Office (DRO), Academic Testing Services wants to ensure that you receive the appropriate accommodations when you go to the test center.

- When scheduling at the test centers, you will see your approved accommodations listed as part of your appointment.
- Students who plan to use their accommodations at an off-campus location should provide the accommodation information to their proctors.

Off-Campus students — Academic Testing Services, will oversee the process of approving an off-campus proctor, sending all exam materials, and receiving any materials from your proctor. Students are responsible for any additional fees associated with off-campus proctoring. To request off-campus proctoring, visit [Off-Campus Testing](#).

- **ProctorU and other similar non-in-person proctors are not allowed for this course.**

If you have any additional questions, contact testing-services@ncsu.edu or call 919.513.1513.

Attendance and Participation

There are generally no attendance or participation requirements other than completing the work in the course (often with a group). Attendance is really done through your completion of course learning materials and your interaction with your fellow students on slack and in completing assignments.

Grading Scale

This course uses the grading scale below:

Low	Letter	High
98 ≤	A+	≤ 100
93 ≤	A	< 97
90 ≤	A-	< 93
88 ≤	B+	< 90
83 ≤	B	< 87
80 ≤	B-	< 83
78 ≤	C+	< 80

Low	Letter	High
73 ≤	C	< 77
70 ≤	C-	< 73
68 ≤	D+	< 70
63 ≤	D	< 67
60 ≤	D-	< 63
0 ≤	F	< 60

Note: This course may not be taken as an **S/U graded course**.

Students who wish to **audit** the course with satisfactory status must register officially for the course and will be required to complete 6 of the 8 homework assignments and complete at least 2 of the 3 projects with a 70% or better on average to receive credit.

University Policies

Students are responsible for reviewing the NC State University Policies, Rules, and Regulations (PRRs), which pertain to their course rights and responsibilities, including, but not limited to:

- [POL 04.25.05](#) Equal Opportunity and Non-Discrimination Policy Statement
 - [REG 02.20.01](#) Academic Accommodations for Students with Disabilities
- [POL 11.35.01](#) Student Conduct
 - [REG 11.35.05](#) Code of Student Conduct
 - [REG 11.35.02](#) Student Discipline Procedures
- [REG 02.20.03](#) Attendance Regulations
- [REG 02.50.03](#) Grades and Grade Point Average
- [REG 02.20.15](#) Credit-Only Courses
- [REG 02.20.04](#) Audits

[REG 08.00.11](#) Online Course Material Host Requirements